

Engineering Ethics: Challenger Case, Whistleblowing & Public Sa

Subject: ENG-ETH - Engineering Ethics & Professional Conduct | Author: National Academy of Engineering / NSPE
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1. Space Shuttle Challenger Investigation & Lessons

Morton Thiokol engineers warned that cold temperatures would compromise O-ring resilience.
Managerial pressure to launch overrode strict technical risk assessments.
Demonstrates the catastrophic risk of normalizing deviance in safety-critical engineering.

2. NSPE Fundamental Canons & Rules of Practice

Canon 1: Engineers shall hold paramount the safety, health, and welfare of the public.
Canon 2: Perform services only in areas of competence.
Canon 3: Issue public statements only in an objective and truthful manner.

3. Algorithmic Accountability & Modern AI Governance

Software and AI engineers must audit autonomous algorithms for bias, security, and safety.
Ethical whistleblowing protocols safeguard public trust in technological infrastructure.